Joint Estimation of Vehicle States with Spatio-Temporal Friction under Different Road Conditions

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***Abstract*—Accurately estimating and predicting tire–road friction, despite the absence of direct sensors, is essential for optimizing safety-critical control strategies and task planning in advanced driver assistance and autonomous driving systems under dynamic road conditions. Recent approaches propose the integration of camera and vehicle dynamics based methods. These hybrid approaches achieve a more robust estimation, but can not retain and recall this information for a location. They are also unable to quantify uncertainty in tire-road friction with respect to dynamic road conditions. This paper proposes a novel scheme for the estimation of tire-road fiction under uncertain road conditions. A 2D map of the environment is combined with a dynamic vehicle model to enable joint estimation of environment and vehicle state, including tire-road friction. A tailored sequential Monte-Carlo scheme allows for online updating of map information and enables recall upon revisiting a location. Results show, that uncertainty in tire-road friction can be reduced during successive trials of emergency breaking. As a result, predictions of breaking distance gain increased accuracy with a reduction RMSE of $4.95\,\%$ on average.**

***Index Terms*—vehicle safety, friction, state estimation.**

I. Introduction

In recent years, there have been increased efforts on the development of advanced driver assistance systems (ADAS) and autonomous driving technologies [1]. To ensure the safety of the driver and other traffic participants it is crucial, that related systems can accurately characterize the traction potential of the vehicles tires. In the case of collision avoidance, this information can be used to optimize control strategies and the system must be capable to quickly react to changing road conditions [2]. If information on tire road friction is available predictively, autonomous vehicles or robots can utilize these prediction during task planning [3].

Methods for estimation of the tire-road friction coefficient are generally classified into cause- and effect-based methods [4] and there are reviews, which provide an extensive comparison of different methods [5]–[7].

Cause-based methods derive information on the tire-road friction from external factors on the tire-road contact condition and can be further divided into methods that target pavement roughness as well as woad-weather condition [7]. Cause-based methods can be used to gather rough information on tire-road friction during most driving situations. Examples include image based classification [8]–[11] or utilization of acoustic data to either estimate surface roughness [12] or type [13]. They generally require extensive training data to yield robust algorithms. Additionally, their accuracy can quickly degrade in situations which where not considered during training and their extrapolation ability may be poor [5], [6], [14].

Effect-based methods infer information on tire-road friction from responses of tire or vehicle to tire-road contact conditions. They can be further divided in methods based on vehicle dynamics [15], [16], tire sensors or slip estimation [17]–[19]. As an alternative to using physical models, data driven [20]–[22] and hybrid approaches [23]–[27] are also possible. Effect-based methods can provide direct information on the tire-road friction condition. However, methods relying on vehicle dynamics or slip generally require high levels of excitation. The tire must reach slip conditions close to its traction maximum at around $30\,\%$ to $50\,\%$ slip. Such conditions are rare in most driving situations. These methods also may struggle with rapid changes in tire-road friction e.g. on change of pavement type [5], [6], [14]. Dynamics based approaches may increase robustness against estimation drift during low excitation phases, by enabling the friction estimation based on local observability [28], [29].

Both groups of methods in isolation may not yield satisfactory results when coupled with adaptive motion control or planning behavior. But given the individual strengths and weaknesses, there is potential to combine different approaches and utilize heterogeneous data to enable more robust estimation of the tire-road friction condition [30]. In recent years, the integration of image based road-surface classification and dynamics based estimators has been proposed by various authors [30]–[33]. Information on the road condition, obtained from images, can be used to determine the plausible value range of tire-road friction. This can be applied to the dynamics based estimation as a reference [31], [33], or to truncate the the final result [31]. If the camera is forward facing, information can be gathered predictively along the cameras field of view [30]. Integrating camera information in this way, can enable a better response to changing road conditions. However, these approaches are reliant on high confidence predictions of the deployed image processing scheme. As mentioned above, unforeseen environmental conditions can degrade this performance. It has also been shown, that the inclusion of additional heterogeneous data can further enhance the road condition estimation [34]. The above integration schemes directly relate image information to tire-road friction, making the inclusion of other heterogeneous data not straightforward. Additionally, the integration with estimators like the unscented Kalman filter, which assumes a Gaussian distribution, makes these algorithms unable to express uncertainties

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with respect to different environmental conditions. While a camera with forward facing field of view enables a certain planning horizon, the information always remains relative to the vehicles current position. The estimation results are not saved and can not be recalled when the vehicle revisits a location at a later time.

In this paper, we present a novel scheme for the estimation of the tire road-friction condition. We propose to represent the causal chain from environmental causes of tire-road friction to the resulting effects on the vehicle dynamics as a state space model. Sequential Monte-Carlo methods are used to jointly estimate environmental and vehicle state, including tire-road friction. In doing so, we are able to quantify uncertainties of tire-road friction with respect to different environmental conditions and propagate this uncertainty through the vehicle dynamics. The environment is represented by a 2D map comprised of discrete road patches. By estimating the vehicles position, a corresponding patch is selected and associated information about the probability densities of environmental states are used during the prediction step. The tailored Monte-Carlo scheme allows for the recursive update of map information from state estimates. This information can be recalled when making new predictions on the corresponding road patch. By explicitly mapping environmental states over 2D space, it is also possible to make predictions independent of the vehicles position. To the best of our knowledge, we achieve for the first time, 2D mapping of tire-road friction in an online setting, by combining cause- and effect-based methods. Interdependence relationships between environmental states can be factorized as a Bayesian network, which is based on our previous work [35], [36]. This definition makes the model more modular, and adaptation to different sensor setups is more straightforward. A simplified illustration of the proposed method is given in Fig. 1.

The rest of the paper is organized as follows. Section II presents the general problem formulation and gives details on road condition and vehicle model. Section IV presents the experimental results. Section V gives a conclusion and an outlook for further research.

Notation: For a vector $\boldsymbol{e} \in \mathbb{R}^{n_e}$, $\boldsymbol{e} \sim \mathcal{N}(\boldsymbol{m}, \boldsymbol{\Sigma})$ denotes a draw from a multivariate Normal with mean $\boldsymbol{m}$ and covariance $\boldsymbol{\Sigma}$, and e_i is its i-th element. We use column vectors if not stated explicitly otherwise. A matrix $\boldsymbol{A}$ is written in bold and capital. If we call matrix $\boldsymbol{A}$ a look-up table, entries a_{ij} for row i and column j are addressed as $\boldsymbol{A}(i, j)$. We write the conditional density of a state sequence $\boldsymbol{x}_{1:t} := \{\boldsymbol{x}_i\}_{i=1}^{t}$ from time steps 1 to t, given the measurements $\boldsymbol{y}_{1:t}$ as $p(\boldsymbol{x}_{1:t}|\boldsymbol{y}_{1:t})$. To avoid excessive use of indices, writing $\boldsymbol{x}_t = [a, b]^{\mathrm{T}}$ implies the same time dependence of $\boldsymbol{x}_t$ in elements a and b. The same holds when writing $p(\boldsymbol{x}_t) = p(a|b)\, p(b)$. The notation $\boldsymbol{\Theta}_{t|t-1}$ refers to the one-step prediction of variable $\boldsymbol{\Theta}$ from time step $t-1$, and $\boldsymbol{\Theta}_{t|t}$ is the posterior variable, after incorporating the evidence of time step t.

II. Problem Formulation

Consider a vehicle as a non-linear dynamical system, whose state at time t is defined as $\boldsymbol{x}_t$. The vehicle dynamics are

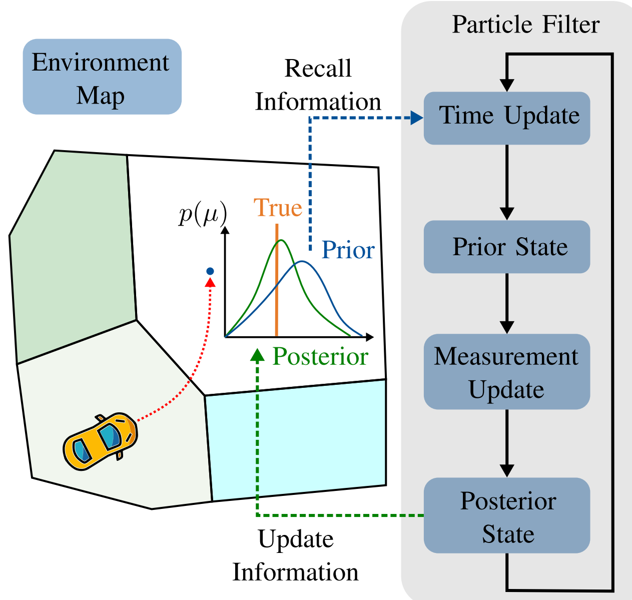

Fig. 1. Simplified illustration of the proposed method. Vehicle and environment state are estimated using a particle filter. Information about the distribution of tire-road friction μ and other environmental states is saved in a map comprised of discrete road patches. Map information is recalled during proposal generation. After incorporating measurements, map information is updated using the posterior state.

subject to local influences of the surrounding environment $\boldsymbol{\Xi}(\boldsymbol{x}_t)$, which could be effects like friction, air current or temperature. These environmental effects are called local, because they depend on the vehicles state by its position $\boldsymbol{p}_t$ as part of $\boldsymbol{x}_t$ at time t. The vehicle's state $\boldsymbol{x}_t$ is observed through the measurements $\boldsymbol{y}_t$, obtained from sensors mounted on the vehicle. As the local environmental influences affect these sensors as well, the measurement function depends on $\boldsymbol{x}_t$ and $\boldsymbol{\Xi}$. The vehicle's dynamics can be described by the nonlinear, time-discrete state-space model (SSM)

$$\begin{aligned} \boldsymbol{x}_t &= \boldsymbol{f}(\boldsymbol{x}_{t-1}, \boldsymbol{\Xi}(\boldsymbol{x}_{t-1})) + \boldsymbol{e}, \\ \boldsymbol{y}_t &= \boldsymbol{h}(\boldsymbol{x}_t, \boldsymbol{\Xi}(\boldsymbol{x}_t)) + \boldsymbol{w}, \end{aligned} \tag{1}$$

with additive, stochastic noise $\boldsymbol{e}$ and $\boldsymbol{w}$. Function $\boldsymbol{f}$ are known system dynamics of the vehicle, which can be derived by common approaches like the single- or two-track model. Function $\boldsymbol{h}$ is the mapping of states to available measurements $\boldsymbol{y}_t$. The vehicle is controlled by a known set of input signals, which are steering angle, break pressure, engine torque and gear transmission. Without loss of generality, the input-dependence is not written explicitly to retain a concise notation. The vehicle state

$$\boldsymbol{x}_t = [\boldsymbol{p}, \psi, \dot{\psi}, \omega_{1:4}, v_x, v_y]^{\mathrm{T}}, \tag{2}$$

is described by 2D Cartesian position vector $\boldsymbol{p}$, yaw angle ψ, tire rotational rates $\omega_{1:4}$ for all four tires and velocities at the center of gravity v_x and v_y in longitudinal and lateral direction, respectively. Measurements are

$$\boldsymbol{y}_t = [\boldsymbol{p}_{\mathrm{GPS}}, \boldsymbol{v}_{\mathrm{GPS}}, \dot{\psi}, a_x, a_y, \omega_{1:4}, S_T, S_W, S_P]^{\mathrm{T}}, \tag{3}$$

which relate to the respective dynamic states of the vehicle, where $\boldsymbol{p}_{\mathrm{GPS}}$ and $\boldsymbol{v}_{\mathrm{GPS}}$ are position and velocity from a GPS-receiver, a_x and a_y are longitudinal and lateral acceleration from an inertial measurement unit. The quantities S_T, S_W and S_P are measurements related to the local environmental conditions, with S_T for air temperature, S_W for road-weather condition and S_P for detection of precipitation. The environment $\Xi(\boldsymbol{x}_t)$ represents a spatio-temporal process whose true dynamics are unknown.

The model structure to combine environment model with vehicle dynamics is described in Section III. Based on the proposed model structure, online inference of unknown parameters and latent system states is outlined in Section III-A and III-B, respectively.

III. Model Structure

To address the aforementioned problem, this section will first introduce the general model structure for environment and vehicle in form of a state space model with a system of model equations.

To represent the environment $\Xi(\boldsymbol{x}_t)$ we define an auxiliary variable $\boldsymbol{\xi}_t$ as a realization of environmental effects for $\boldsymbol{x}_t$, which are relevant for the vehicle dynamics. The local environmental effects are defined as

$$\boldsymbol{\xi}_t = [T, P, W, R, \mu]^{\mathrm{T}}, \tag{4}$$

with the pavement temperature T, precipitation P, road-weather condition W, road-pavement type R and the maximum tire-road friction potential μ. Variables T, P, W and R have discrete values, the maximum tire-road friction potential μ is continuous and positive. The discrete variables T, P, W and R can each have K_* possible values, where $*$ represents the respective variable. For example, pavement type R differentiates between *asphalt*, *concrete* and *cobblestone*, precipitation P can be *true* or *false*. Definitions for all variables are given in the Appendix. These environmental effects $\boldsymbol{\xi}_t$ represent the state of the environment at time t and their true values are unknown.

Because pavement temperature T, precipitation P, road-weather condition W and road-pavement type R have discrete values, they can be represented as categorical random variables. To formulate the state space model as a system of equations, categorical variables can be represented by an equivalent functional form. We define a function C, which produces discrete values $k \in \{1, \dots, K\}$ with K possible outcomes by

$$\begin{aligned} &C(\boldsymbol{b}, u) = k \text{ if } \sum_{i=1}^{k-1} b_i \le u < \sum_{i=1}^{k} b_i, \\ &\text{with } u \sim \mathcal{U}(0,1), \text{ and } \sum_{i=1}^{K} b_i = 1,\ b_i > 0\,\forall\, i \end{aligned} \tag{5}$$

with parameters $\boldsymbol{b} \in \mathbb{R}^K$ as the probabilities for each class k and u as uniformly distributed process noise. The uniform distribution $\mathcal{U}(0,1)$ is defined in the interval $(0, 1]$. Because tire-road friction μ is always positive, it is more convenient to define this variable as a process over the exp-operator by

$$\begin{aligned} \mu &= \exp(m_\mu + \epsilon), \\ \epsilon &\sim \mathcal{N}(0, \sigma_\mu), \end{aligned} \tag{6}$$

with exponents m_μ and ϵ, and variance σ_μ. With this definition, it can be said that tire-road friction μ follows a log-normal distribution.

The environment model receives its spatial resolution by segmenting the road into discrete patches j. For a given location $\boldsymbol{p}_t$, it is possible to deterministically select the corresponding patch j. Using this segmentation, a different set of parameters $\boldsymbol{b}$, m_μ and σ_μ can be used to describe the environment model for each patch j separately, e.g., based on prior knowledge from other sources such as weather forecasts, some patches are more likely to have precipitation compared to others. Additionally, the environment variables in $\boldsymbol{\xi}_t$ are connected by interdependence relationships, which can be modeled using Bayesian networks, as done in [35], [36]. The road-weather condition W is influenced by precipitation P and pavement temperature T. The tire-road friction μ results from a combination of pavement type R and road-weather condition W. Since all conditional variables are discrete, we can define separate parameter combinations for these conditionals. For

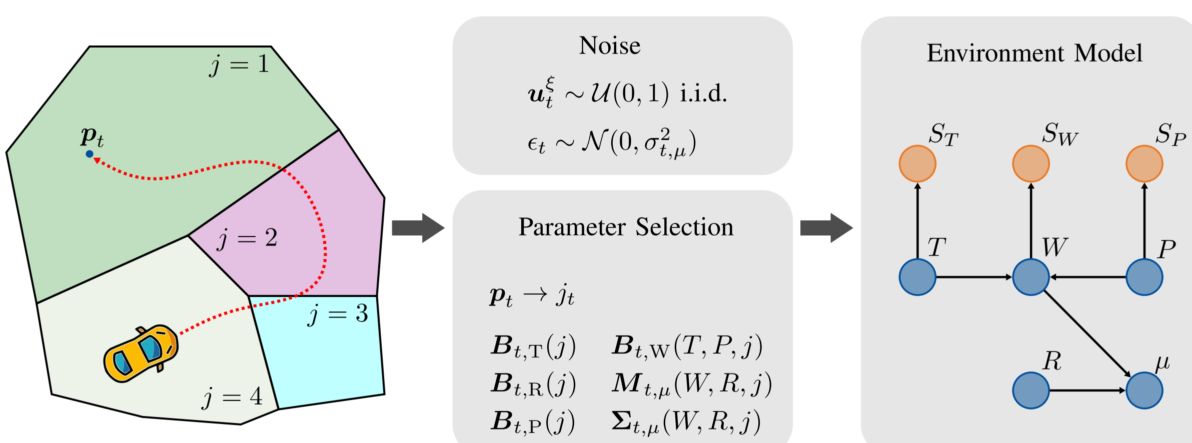

Fig. 2. Illustration of the environment model. The model gains a spatial resolution by selecting the nodes prior based on the proposals for the vehicles position $\boldsymbol{p}_t$. The space is represented by discrete patches j, each associated with their own set of parameters. Information about the environment is gathered by discrete measurements S_T, S_W and S_P. State variables are represented in blue, sensors in orange.

example, we have a different exponent m_μ and variance σ_μ for each combination of pavement type P and road weather condition W. To consider all possible combinations of environment model parameters, all parameters are grouped into a set of look-up tables

$$\boldsymbol{\theta}_t = \{\boldsymbol{B}_T, \boldsymbol{B}_P, \boldsymbol{B}_W, \boldsymbol{B}_R, \boldsymbol{M}_\mu, \boldsymbol{\Sigma}_\mu\}, \tag{7}$$

where $\boldsymbol{B}_*$ are tables containing conditional probabilities $\boldsymbol{b}_*$ for T, P, W, and R, respectively and $\boldsymbol{M}_\mu$ and $\boldsymbol{\Sigma}_\mu$ are tables containing conditional exponent m_μ and variance σ_μ. The look-up tables in $\boldsymbol{\theta}_t$ are the *unknown* parameters of the environment model, and they can also vary with time t to capture temporal changes in the environment. We address an entry in these look-up tables by a function-like notation, e.g., we denote $\boldsymbol{b}_{T,j} := \boldsymbol{B}_T(j)$ as the vector of conditional probabilities $\boldsymbol{b}_T \in \mathbb{R}^{K_T}$ for pavement temperature T given patch j and $\boldsymbol{B}_T = [\boldsymbol{b}_{T,j}]_{j=1}^{K_J} \in \mathbb{R}^{K_T \times K_J}$. The model structure is illustrated in Fig. 2.

With the above definitions, we can formulate a parametrized function $\hat{\boldsymbol{\Xi}}$ as an approximation of the environment $\boldsymbol{\Xi}$ by a system of equations

$$\begin{aligned}\boldsymbol{\xi}_t &= \hat{\boldsymbol{\Xi}}(\boldsymbol{x}_t, \boldsymbol{\theta}_t, \boldsymbol{u}_t^\xi, \epsilon_t) \\ &= \begin{bmatrix} T \\ P \\ W \\ R \\ \mu \end{bmatrix} = \begin{bmatrix} C(\boldsymbol{B}_T(j), u_1^\xi) \\ C(\boldsymbol{B}_P(j), u_2^\xi) \\ C(\boldsymbol{B}_W(T, P, j), u_3^\xi) \\ C(\boldsymbol{B}_R(j), u_4^\xi) \\ \exp\left(\boldsymbol{M}_\mu(W, R, j) + \epsilon_t\right) \end{bmatrix},\end{aligned} \tag{8}$$

with $\boldsymbol{u}_t^\xi \sim \mathcal{U}(0,1)$ independent and identically distributed (i.i.d.) process noise and unknown set of parameters $\boldsymbol{\theta}_t$. Information about the environment at time t is gathered by discrete measurements S_T, S_W and S_P. Given T, W and P, these measurements are independent of the vehicle's state $\boldsymbol{x}_t$. Due to this independence, we split up the measurement model into two parts and define the measurements as $\boldsymbol{y}_t = \{\boldsymbol{y}_t^x, \boldsymbol{y}_t^\xi\}$. First, we have measurements which relate to the vehicles dynamic state

$$\begin{aligned}\boldsymbol{y}_t^x &= [\boldsymbol{p}_{\text{GPS}}, \boldsymbol{v}_{\text{GPS}}, \dot{\psi}, a_x, a_y, \omega_{1:4}]^{\text{T}}, \\ &= \boldsymbol{h}_x(\boldsymbol{x}_t, \mu_t) + \boldsymbol{w},\end{aligned} \tag{9}$$

with zero-mean, additive Gaussian noise $\boldsymbol{w} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{R})$ and known covariance $\boldsymbol{R}$. Second, we have measurements of the environment

$$\begin{aligned}\boldsymbol{y}_t^\xi &= [S_T, S_W, S_P]^{\text{T}}, \\ &= \boldsymbol{h}_\xi(\boldsymbol{\xi}_t, \boldsymbol{u}_t^y) = \begin{bmatrix} C(\boldsymbol{B}_{S_T}(T), u_1^y) \\ C(\boldsymbol{B}_{S_W}(W), u_2^y) \\ C(\boldsymbol{B}_{S_P}(P), u_3^y) \end{bmatrix}\end{aligned} \tag{10}$$

with i.i.d. $\boldsymbol{u}_t^y \sim \mathcal{U}(0,1)$ as measurement noise and known conditional probability tables $\boldsymbol{B}_{S_T}$, $\boldsymbol{B}_{S_W}$ and $\boldsymbol{B}_{S_P}$ equivalent to (5).

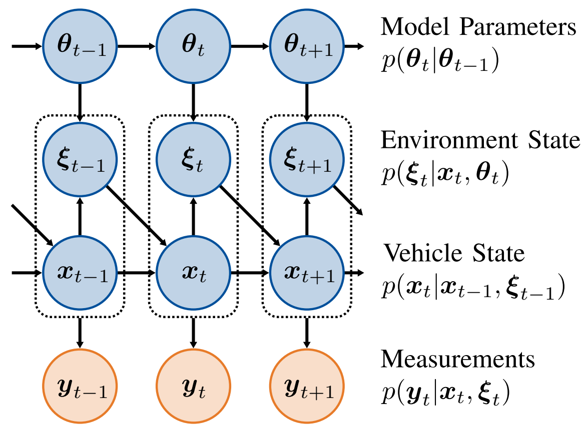

Fig. 3. Model structure of the joint state space, represented as a Markov-chain. The vehicle state moves from $\boldsymbol{x}_{t-1}$ to $\boldsymbol{x}_t$ under influence of the environment $\boldsymbol{\xi}_{t-1}$ at time $t-1$. The state of the environment $\boldsymbol{\xi}_t$ at time t as experienced by the vehicle then depends on the vehicles position via $\boldsymbol{x}_t$ and the model parameters $\boldsymbol{\theta}_t$. Nodes of latent variables are colored blue, measurements are colored orange.

The entire system can be summarized by the following set of equations

$$\begin{aligned}\boldsymbol{x}_t &= \boldsymbol{f}(\boldsymbol{x}_{t-1}, \mu_{t-1}) + \boldsymbol{e}, \quad & \mu_{t-1} \in \boldsymbol{\xi}_{t-1}, \\ \boldsymbol{\xi}_t &= \hat{\boldsymbol{\Xi}}(\boldsymbol{p}_t, \boldsymbol{\theta}_t, \boldsymbol{u}_t^\xi, \epsilon_t), & \boldsymbol{p}_t \in \boldsymbol{x}_t, \\ \boldsymbol{y}_t^x &= \boldsymbol{h}_x(\boldsymbol{x}_t, \mu_t) + \boldsymbol{w}, \\ \boldsymbol{y}_t^\xi &= \boldsymbol{h}_\xi(\boldsymbol{\xi}_t, \boldsymbol{u}_t^y),\end{aligned} \tag{11}$$

where $\boldsymbol{e} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{Q})$ is zero-mean, additive Gaussian noise with known covariance matrix $\boldsymbol{Q}$. The vehicle dynamics are subject to influences of the environment $\boldsymbol{\xi}_{t-1}$ in terms of the maximum tire-road friction potential μ_{t-1}, which itself depends on a combination of $[T_{t-1}, P_{t-1}, W_{t-1}, R_{t-1}]$. The system is visualized as a Markov-chain in Fig. 3.

Goal is the joint estimation of latent state trajectories $\boldsymbol{x}_{0:t}$, $\boldsymbol{\xi}_{0:t}$ and parameters $\boldsymbol{\theta}_t$ given measurements $\boldsymbol{y}_{0:t}$ at time t. With the given model structure, the likelihood $p(\boldsymbol{y}_t|\boldsymbol{x}_t, \boldsymbol{\xi}_t)$ is independent of the model parameters $\boldsymbol{\theta}_t$. As a result, the filtering posterior can be factorized by

$$\begin{aligned}p(\boldsymbol{x}_{0:t}, \boldsymbol{\xi}_{0:t}, \boldsymbol{\theta}_t|\boldsymbol{y}_{0:t}) = p(\boldsymbol{\theta}_t&|\boldsymbol{x}_{0:t}, \boldsymbol{\xi}_{0:t}) \\ &p(\boldsymbol{x}_{0:t}, \boldsymbol{\xi}_{0:t}|\boldsymbol{y}_{0:t}).\end{aligned} \tag{12}$$

The second term on the right hand side $p(\boldsymbol{x}_{0:t}, \boldsymbol{\xi}_{0:t}|\boldsymbol{y}_{0:t})$ can be approximated using tailored sequential Monte-Carlo (SMC) methods. The SMC algorithm provides proposals for state trajectories $\boldsymbol{x}_{0:t}$ and $\boldsymbol{\xi}_{0:t}$. Choosing a conjugate prior for parameters $\boldsymbol{\theta}_t$ enables to compute a closed form solution of parameter posterior $p(\boldsymbol{\theta}_t|\boldsymbol{x}_{0:t}, \boldsymbol{\xi}_{0:t})$.

The proposed SMC scheme is illustrated in Fig. 4. Subsequent Sections III-A and III-B will provide details on the estimation of parameter and state posterior respectively. The complete algorithm is summarized in Section III-C.

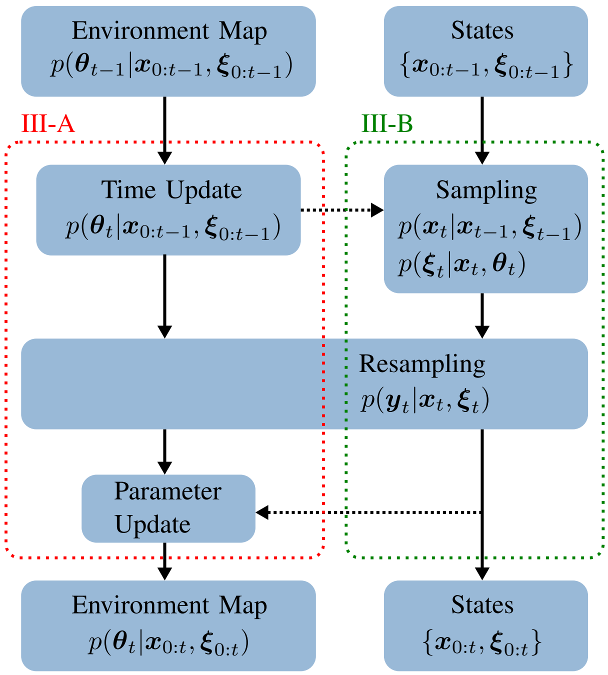

Fig. 4. Illustration of the proposed SMC scheme based on the model structure in (11).

A. *Estimating the parameter posterior*

Based on the model structure in (11), the parameter posterior $p(\boldsymbol{\theta}_t|\boldsymbol{x}_{0:t},\boldsymbol{\xi}_{0:t})$ at time t can be decomposed as

$$\begin{aligned} p(\boldsymbol{\theta}_t|\boldsymbol{x}_{0:t},\boldsymbol{\xi}_{0:t}) \propto\ & p(\boldsymbol{\xi}_t|\boldsymbol{x}_t,\boldsymbol{\theta}_t) \\ & p(\boldsymbol{\theta}_t|\boldsymbol{x}_{0:t-1},\boldsymbol{\xi}_{0:t-1}), \end{aligned} \tag{13}$$

using Bayes' rule, where $p(\boldsymbol{\theta}_t|\boldsymbol{x}_{0:t-1},\boldsymbol{\xi}_{0:t-1})$ is the current parameter prior and $p(\boldsymbol{\xi}_t|\boldsymbol{x}_t,\boldsymbol{\theta}_t)$ its likelihood, which are introduced subsequently. Including the environmental models topology from (8) and the definition of $\boldsymbol{\theta}$ in (7) allows for further factorization of the likelihood as

$$\begin{aligned} p(\boldsymbol{\xi}_t|\boldsymbol{x}_t,\boldsymbol{\theta}_t) = p(\mu|&\boldsymbol{M}_\mu(W,R,j),\boldsymbol{\Sigma}_\mu(W,R,j)) \\ & p(W|\boldsymbol{B}_W(T,P,j)) \\ & p(T|\boldsymbol{B}_T(j))\ p(P|\boldsymbol{B}_P(j)) \\ & p(R|\boldsymbol{B}_R(j)), \end{aligned} \tag{14}$$

where the spatial location is represented as the index of road patch j. Choosing a conjugate prior over parameters $\boldsymbol{\theta}_t$ enables the recursive calculation of the parameter posterior by summation of the previous sufficient statistics of $\{\boldsymbol{x}_{0:t-1},\boldsymbol{\xi}_{0:t-1}\}$ and the new statistics of $\boldsymbol{x}_t$ and $\boldsymbol{\xi}_t$. By exploiting the hierarchical structure in (14) we can define a conjugate prior for the unknown parameters of each term *separately*.

We first introduce a general definition of conjugate priors based on [37]. Some random variable $\boldsymbol{\xi} \sim p(\boldsymbol{\xi}|\boldsymbol{\theta})$ may follow a distribution which depends on some unknown parameters $\boldsymbol{\theta}$. These unknown parameters $\boldsymbol{\theta}$ can be inferred from random samples of $\boldsymbol{\xi}$ using Bayes rule. This requires the definition of a prior $p(\boldsymbol{\theta}|\boldsymbol{\Theta})$ over these parameters, whose functional form is shaped by its set of hyper-parameters $\boldsymbol{\Theta}$, e.g. mean and variance for the normal distribution. Given samples of $\boldsymbol{\xi}$, the posterior can be calculated using Bayes rule with

$$p(\boldsymbol{\theta}|\boldsymbol{\Theta},\boldsymbol{\xi}) \propto p(\boldsymbol{\xi}|\boldsymbol{\theta})\ p(\boldsymbol{\theta}|\boldsymbol{\Theta}) \tag{15}$$

In this constellation, $p(\boldsymbol{\xi}|\boldsymbol{\theta})$ is the parameter likelihood and $p(\boldsymbol{\theta}|\boldsymbol{\Theta})$ the parameter prior. If the prior $p(\boldsymbol{\theta}|\boldsymbol{\Theta})$ is conjugate to the likelihood $p(\boldsymbol{\xi}|\boldsymbol{\theta})$, the posterior $p(\boldsymbol{\theta}|\boldsymbol{\Theta},\boldsymbol{\xi})$ has the same functional form as the prior. This means, that it is described by the same kind of parameters $\boldsymbol{\Theta}$ as the prior $p(\boldsymbol{\theta}|\boldsymbol{\Theta})$. As a result, calculation of the posterior $p(\boldsymbol{\theta}|\boldsymbol{\Theta},\boldsymbol{\xi})$ can be done analytically by updating hyper-parameters $\boldsymbol{\Theta}$ given data $\boldsymbol{\xi}$. When the hyper-parameters $\boldsymbol{\Theta}$ are expressed in terms of sufficient statistics, the update can be expressed as the sum of prior sufficient statistics $\boldsymbol{\Theta}_0$ and data sufficient statistics $\boldsymbol{\Theta}(\boldsymbol{\xi})$. Note that in further notation we avoid explicitly expressing this sum, when mentioning distributions $p(\boldsymbol{\theta}|\boldsymbol{\Theta})$ to keep the notation concise. The form of sufficient statistics is derived from the likelihood's probability density or mass function. Combinations of likelihoods and conjugate priors can be found in the literature [38].

With the definition of C in (5), the discrete random variables T, P, W and R follow a categorical distribution $k \sim \mathcal{C}(k|\boldsymbol{b})$, with unknown probabilities $\boldsymbol{b}$ for their classes. For this distribution, the conjugate prior is a Dirichlet-distribution

$$\begin{aligned} \boldsymbol{b} &\sim \mathcal{D}(\boldsymbol{b}|\boldsymbol{z}) \\ \mathcal{D}(\boldsymbol{b}|\boldsymbol{z}) &= \frac{1}{B(\boldsymbol{z})} \prod_{i=1}^{K} b_i^{z_i - 1} \end{aligned} \tag{16}$$

with multivariate beta function B and hyper-parameter $\boldsymbol{z} \in \mathbb{R}^K$, which contains the number of occurrences for observed classes i [37].

With the definition for tire-road friction μ in (8), the variable follows a log-normal distribution, where the exponent is of unknown mean m_μ and variance σ_μ^2. By defining the random variable over the space of $\log \mu$, the conjugate prior is the normal-inverse-gamma distribution

$$\begin{aligned} \log \mu &\sim \mathcal{N}(\log \mu | m, \sigma^2) \\ m, \sigma^2 &\sim \mathcal{NIG}(m, \sigma^2 | \bar{m}, \lambda, \alpha, \beta) \\ &= \mathcal{N}(m|\bar{m}, \sigma^2/\lambda)\ \mathcal{IG}(\sigma^2|\alpha,\beta), \end{aligned} \tag{17}$$

with hyper-parameters mean $\bar{m}$, effective sample size λ, shape α and scale β [38]. These hyper-parameters can be expressed as sufficient statistics by the mapping

$$\begin{aligned} \chi_1 &= \alpha, & \chi_2 &= \lambda \bar{m}, \\ \chi_3 &= \lambda, & \chi_4 &= 2\beta + \lambda \bar{m}^2, \end{aligned} \tag{18}$$

and the inverse mapping

$$\begin{aligned} \alpha &= \chi_1, & \bar{m} &= \frac{\chi_2}{\chi_3}, \\ \lambda &= \chi_3, & \beta &= \frac{1}{2}\left(\chi_4 - \frac{\chi_2^2}{\chi_3}\right), \end{aligned} \tag{19}$$

where we define the grouping $\boldsymbol{\chi}_\mu = \{\chi_1, \chi_3, \chi_3, \chi_4\}$.

Due to the interdependence between the environment variables in $\boldsymbol{\xi}$, the sets of hyper-parameters are grouped into look-up tables, analogous to $\boldsymbol{\theta}$ in (7). An overview for the parameters, their likelihoods and conjugate priors is given in

Table I. The total set of hyper-parameters for the parameter prior $p(\boldsymbol{\theta}_t|\boldsymbol{x}_{0:t-1}, \boldsymbol{\xi}_{0:t-1})$ are the sufficient statistics

$$\boldsymbol{\Theta} = \{\boldsymbol{Z}_T, \boldsymbol{Z}_P, \boldsymbol{Z}_W, \boldsymbol{Z}_R, \boldsymbol{X}_\mu\}. \tag{20}$$

As previously in (7), a vector of sufficient statistics is denoted by a function-like notation, e.g., $\boldsymbol{z}_{T,j} := \boldsymbol{Z}_T(j)$ and $\boldsymbol{Z}_T = [\boldsymbol{z}_{T,j}]_{j=1}^{K_J} \in \mathbb{R}^{K_T \times K_J}$.

TABLE I
OVERVIEW FOR LIKELIEHOOD AND CONJUGATE PRIORS WITH THEIR RESPECTIVE PARAMETERS FOR THE ENVIRONMENT VARIABLES.

Variable	Likelihood	Parameter	Prior	Hyper-Parameter
T	Categorical	$\boldsymbol{B}_T$	Dirichlet	$\boldsymbol{Z}_T$
P	Categorical	$\boldsymbol{B}_P$	Dirichlet	$\boldsymbol{Z}_P$
W	Categorical	$\boldsymbol{B}_W$	Dirichlet	$\boldsymbol{Z}_W$
R	Categorical	$\boldsymbol{B}_R$	Dirichlet	$\boldsymbol{Z}_R$
μ	log-normal	$\boldsymbol{M}_\mu$, $\boldsymbol{\Sigma}_\mu$	normal-inverse-gamma	$\boldsymbol{X}_\mu$

Because we have chosen conjugate priors, the distributions of current parameter prior $p(\boldsymbol{\theta}_t|\boldsymbol{x}_{0:t-1}, \boldsymbol{\xi}_{0:t-1})$ and posterior $p(\boldsymbol{\theta}_t|\boldsymbol{x}_{0:t}, \boldsymbol{\xi}_{0:t})$ have the same functional form, and can respectively be expressed by

$$\begin{aligned} p(\boldsymbol{\theta}_t|\boldsymbol{x}_{0:t-1}, \boldsymbol{\xi}_{0:t-1}) &:= p(\boldsymbol{\theta}_t|\boldsymbol{\Theta}_{t|t-1}) \\ p(\boldsymbol{\theta}_t|\boldsymbol{x}_{0:t}, \boldsymbol{\xi}_{0:t}) &:= p(\boldsymbol{\theta}_t|\boldsymbol{\Theta}_{t|t}) \end{aligned} \tag{21}$$

with the set of sufficient statistics $\boldsymbol{\Theta}$. Due to the factorization of the environment model in (14), calculation of sufficient statistics is done for each entry separately, given the corresponding dependent variables. With given environmental state $\boldsymbol{\xi}_t$ and patch index j, the sufficient statistics can be updated recursively by

$$\begin{aligned} \boldsymbol{Z}_{t|t,T}(T,j) &= \boldsymbol{Z}_{t|t-1,T}(T,j) + 1, \\ \boldsymbol{Z}_{t|t,P}(P,j) &= \boldsymbol{Z}_{t|t-1,P}(P,j) + 1, \\ \boldsymbol{Z}_{t|t,W}(W,T,P,j) &= \boldsymbol{Z}_{t|t-1,W}(W,T,P,j) + 1, \\ \boldsymbol{Z}_{t|t,R}(R,j) &= \boldsymbol{Z}_{t|t-1,R}(R,j) + 1, \\ \boldsymbol{X}_{t|t,1}(W,R,j) &= \boldsymbol{X}_{t|t-1,1}(W,R,j) + 0.5, \\ \boldsymbol{X}_{t|t,2}(W,R,j) &= \boldsymbol{X}_{t|t-1,2}(W,R,j) + \log\mu, \\ \boldsymbol{X}_{t|t,3}(W,R,j) &= \boldsymbol{X}_{t|t-1,3}(W,R,j) + 1, \\ \boldsymbol{X}_{t|t,4}(W,R,j) &= \boldsymbol{X}_{t|t-1,4}(W,R,j) + (\log\mu)^2. \end{aligned} \tag{22}$$

The change of parameters over time is modeled by applying exponential forgetting to the data sufficient statistics. In doing so, less weight is given to older estimates of vehicle state and environment. With exponential forgetting, the time update step $p(\boldsymbol{\theta}_t|\boldsymbol{\theta}_{t-1})$ of the data sufficient statistics is

$$\boldsymbol{\Theta}_{t|t-1} = \gamma \cdot \boldsymbol{\Theta}_{t-1|t-1}, \tag{23}$$

with $0 \le \gamma \le 1$ as the forgetting factor.

B. Estimating the latent states

The posterior for latent system states $\boldsymbol{x}_{0:t}$ and $\boldsymbol{\xi}_{0:t}$ in (12) is approximated by

$$p(\boldsymbol{x}_{0:t}, \boldsymbol{\xi}_{0:t}|\boldsymbol{y}_{0:t}) \approx \sum_{i=1}^{N} q_t^i \delta_{\{\boldsymbol{x}_{0:t}^i, \boldsymbol{\xi}_{0:t}^i\}}(\boldsymbol{x}_{0:t}, \boldsymbol{\xi}_{0:t}) \tag{24}$$

as a set of N weighted particles with importance weights q_t^i. These trajectories are generated using sequential Monte-Carlo methods by recursively sampling new proposals $\{\boldsymbol{x}_t, \boldsymbol{\xi}_t\}$ from a proposal distribution $\pi(\boldsymbol{x}_t, \boldsymbol{\xi}_t|\boldsymbol{x}_{0:t-1}, \boldsymbol{\xi}_{0:t-1}, \boldsymbol{y}_{0:t})$. Importance weights $\{q_t^i\}_{i=1}^N$ are be updated by

$$q_t^i \propto \frac{p(\boldsymbol{y}_t|\boldsymbol{x}_t^i, \boldsymbol{\xi}_t^i)\, p(\boldsymbol{x}_t^i, \boldsymbol{\xi}_t^i|\boldsymbol{x}_{t-1}^i, \boldsymbol{\xi}_{t-1}^i)}{\pi(\boldsymbol{x}_t^i, \boldsymbol{\xi}_t^i|\boldsymbol{x}_{t-1}^i, \boldsymbol{\xi}_{t-1}^i, \boldsymbol{y}_{0:t})}\, q_{t-1}^i, \tag{25}$$

with q_{t-1}^i being the importance weights from the previous time step. To define the sequential Monte-Carlo scheme, we first have to design a proposal distribution $\pi(\boldsymbol{x}_t, \boldsymbol{\xi}_t|\boldsymbol{x}_{0:t-1}, \boldsymbol{\xi}_{0:t-1}, \boldsymbol{y}_{0:t})$. Afterwards we can formulate the update law for importance weights q_t^i.

A common choice in the literature is to directly use the models transitional kernel, in this case $p(\boldsymbol{x}_t, \boldsymbol{\xi}_t|\boldsymbol{x}_{t-1}, \boldsymbol{\xi}_{t-1})$, as proposal distribution. This is known as bootstrap particle filtering. In this work we can exploit the environmental models structure to sample from the optimal proposal distribution for pavement temperature T, precipitation P and road-weather condition W. Additionally, by marginalizing over $\boldsymbol{\theta}_t$ we can sample proposals $\{\boldsymbol{x}_t, \boldsymbol{\xi}_t\}$ directly. To this end, we expand

$$\begin{aligned} &p(\boldsymbol{x}_t, \boldsymbol{\xi}_t|\boldsymbol{x}_{t-1}, \boldsymbol{\xi}_{t-1}) \\ &\quad = p(\boldsymbol{x}_t|\boldsymbol{x}_{t-1}, \boldsymbol{\xi}_{t-1})\, p(\boldsymbol{\xi}_t|\boldsymbol{x}_t, \boldsymbol{\Theta}_t) \\ &\quad = p(\boldsymbol{x}_t|\boldsymbol{x}_{t-1}, \boldsymbol{\xi}_{t-1}) \int p(\boldsymbol{\xi}_t|\boldsymbol{x}_t, \boldsymbol{\theta}) p(\boldsymbol{\theta}|\boldsymbol{\Theta}_t)\, d\boldsymbol{\theta}, \end{aligned} \tag{26}$$

where the integrant contains the hierarchical model of the environment from (14) and the parameter prior from (21). The integral in (26) can be solved analytically by calculating the respective conjugate priors predictive distribution for all environmental states in $\boldsymbol{\xi}_t$. For the discrete random variables T, P, W and R the conjugate prior is a Dirichlet-distribution as shown in (16). The corresponding predictive distribution is a categorical distribution with

$$k \sim \mathcal{C}(k|\tilde{\boldsymbol{z}}_k),\ \tilde{\boldsymbol{z}}_k = \frac{\boldsymbol{z}_k}{||\boldsymbol{z}_k||_1} \tag{27}$$

where k is a drop-in replacement for T, P, W and R. For the normal-inverse-gamma distribution in (17), the predictive distribution is the Student-t distribution with

$$\log\mu \sim \mathcal{T}(\log\mu|\bar{m}, \frac{\beta(\nu+1)}{\nu\alpha}, 2\alpha), \tag{28}$$

with parameters for location, scale and degrees of freedom respectively [38]. Using these predictive distributions, we have an equivalent representation of (14) in terms of the sufficient statistics $\boldsymbol{\Theta}_t$ as

$$\begin{aligned} p(\boldsymbol{\xi}_t|\boldsymbol{x}_t, \boldsymbol{\Theta}_t) = {} & p(\mu|\boldsymbol{X}_\mu(W,R,j)) \\ & p(W|\boldsymbol{Z}_W(T,P,j)) \\ & p(T|\boldsymbol{Z}_T(j))\, p(P|\boldsymbol{Z}_P(j)) \\ & p(R|\boldsymbol{Z}_R(j)). \end{aligned} \tag{29}$$

Using (26) with (29) as proposal distribution would yield the bootstrap particle filter. The efficiency of proposal generation can be increased by incorporating measurements $\boldsymbol{y}_t$ into the sampling process. With the structure of the environmental model shown in Fig. (2), this can be done analytically for measurements $\boldsymbol{y}_t^\xi = [S_T, S_P, S_W]^\mathrm{T}$. The system definition in (11) allows a factorization of the state likelihood as

$$p(\boldsymbol{y}_t|\boldsymbol{x}_t,\boldsymbol{\xi}_t) = p(\boldsymbol{y}_t^x|\boldsymbol{x}_t,\mu_t)\; p(\boldsymbol{y}_t^\xi|\boldsymbol{\xi}_t). \tag{30}$$

With measurements $\boldsymbol{y}_t^\xi$ being also discrete variables, the respective posterior distribution for T, P and W can be calculated in a similar manner to variable elimination in a standard discrete Bayesian network [39]. The posterior predictive distributions are calculated by

$$\begin{aligned} p(W|\boldsymbol{Z}_W(T,P,j),S_W) &\propto \tilde{\boldsymbol{Z}}_W(T,P,j)\;\cdot \boldsymbol{B}_{S_W}(S_W),\\ p(T|\boldsymbol{Z}_T(j),S_T) &\propto \tilde{\boldsymbol{Z}}_T(j)\;\cdot \boldsymbol{B}_{S_T}(S_T),\\ p(P|\boldsymbol{Z}_P(j),S_P) &\propto \tilde{\boldsymbol{Z}}_P(j)\;\cdot \boldsymbol{B}_{S_P}(S_P). \end{aligned} \tag{31}$$

Using (31), we can the fine the posterior predictive distribution for environment states $\boldsymbol{\xi}_t$ given measurements $\boldsymbol{y}_t^\xi$ by

$$\begin{aligned} p(\boldsymbol{\xi}_t|\boldsymbol{x}_t,\boldsymbol{\Theta}_t,\boldsymbol{y}_t^\xi) &\propto p(\boldsymbol{y}_t^\xi|\boldsymbol{\xi}_t)\; p(\boldsymbol{\xi}_t|\boldsymbol{x}_t,\boldsymbol{\Theta}_t)\\ &\propto p(\mu|\boldsymbol{X}_\mu(W,R,j))\\ &\quad p(W|\boldsymbol{Z}_W(T,P,j),S_W)\\ &\quad p(T|\boldsymbol{Z}_T(j),S_T)\\ &\quad p(P|\boldsymbol{Z}_P(j),S_P)\\ &\quad p(R|\boldsymbol{Z}_R(j)). \end{aligned} \tag{32}$$

Using (26) and (32), we can define the proposal distribution as

$$\begin{aligned} \pi(\boldsymbol{x}_t,\boldsymbol{\xi}_t|\boldsymbol{x}_{t-1}^i,\boldsymbol{\xi}_{t-1}^i,\boldsymbol{y}_{0:t}) &= p(\boldsymbol{x}_t|\boldsymbol{x}_{t-1}^i,\boldsymbol{\xi}_{t-1}^i)\\ &\quad p(\boldsymbol{\xi}_t|\boldsymbol{x}_t,\boldsymbol{\Theta}_t,\boldsymbol{y}_t^\xi) \end{aligned} \tag{33}$$

Inserting (29), (26), (30) and (33) in (25) yields the new update for the importance weights

$$q_t^i \propto p(\boldsymbol{y}_t^x|\boldsymbol{x}_t^i,\mu_t^i). \tag{34}$$

Note that the update of the importance weights now only depends on measurements $\boldsymbol{y}_t^x$ and all other terms in the expansion of (25) cancel. The weights q_{t-1}^i of the previous time step are canceled by the resampling process.

To summarize, the particle filter operates by first sampling new proposals for $\{\boldsymbol{x}_t^i,\boldsymbol{\xi}_t^i\}$ from the proposal distribution in (33), which are then resampled using the importance weights from (34). This yields samples of the target distribution $p(\boldsymbol{x}_{0:t},\boldsymbol{\xi}_{0:t}|\boldsymbol{y}_{0:t})$, which are used to update the hyperparameters $\boldsymbol{\Theta}$ as described in Section III-A.

C. Complete algorithm

The complete algorithm is summarized in Algorithm 1. The parameter prior for $t=0$ is initialized by setting values for prior sufficient statistics $\boldsymbol{\Theta}_0$, the data sufficient statistics $\boldsymbol{\Theta}_{0|0}^i$ are initialized with zero for each sample i. Details for prior parametrization of the environment model are provided in the Appendix. The time update by exponential forgetting in line 2 of the algorithm is only applied to the data sufficient statistics $\boldsymbol{\Theta}_{t-1|t-1}^i$. In doing so, the parameter posterior will asymptotically re-approach the prior when no data is collected over time. New particles for $\boldsymbol{\xi}_t^i$ can be generated via Gibbs sampling from the factors of the proposal distribution in (33) and (32). To generate proposals for tire-road friction μ, this also involves the inverse transformation of sufficient statistics $\boldsymbol{X}_{t,\mu}(R_t,W_t,j_t)$ by (19). In implementation, variables for sufficient statistics are simply multidimensional matrices. Because conditional variables like pavement temperature T or precipitation P are discrete classes, they are represented by integers and can be directly used as indices. The update of the data sufficient statistics $\boldsymbol{\Theta}_{t|t-1}^i$ according to (22) is therefore only performed on individual entries of these matrices.

Algorithm 1 Marginalized particle filter for joint estimation of latent environmental and vehicle states

Initialize: Data $\boldsymbol{y}_{0:T}$, parameter prior $p(\boldsymbol{\theta})$, particles $\{\boldsymbol{x}_0^i,\boldsymbol{\xi}_0^i\}_{i=1}^N \sim p(\boldsymbol{x}_0,\boldsymbol{\xi}_0|\boldsymbol{\theta})$, weights $\{q_0^i\}_{i=1}^N = 1/N$.

1: **for** $t = 1,\ldots,T$ **do**
2: Statistics time update $\boldsymbol{\Theta}_{t|t-1}^i \leftarrow \boldsymbol{\Theta}_{t-1|t-1}^i$. ▷ by (23)
3: Draw $\boldsymbol{x}_t^i \sim \mathcal{N}(\boldsymbol{x}_t|\boldsymbol{f}(\boldsymbol{x}_{t-1}^i,\boldsymbol{\xi}_{t-1}^i),\boldsymbol{Q})$.
4: Determine patches j^i from $\boldsymbol{x}_t^i$.
5: Draw $\boldsymbol{\xi}_t^i \sim p(\boldsymbol{\xi}_t|\boldsymbol{x}_t^i,\boldsymbol{\Theta}_{t|t-1}^i)$. ▷ by (32)
6: Statistics measurement update $\boldsymbol{\Theta}_{t|t}^i \leftarrow \boldsymbol{\Theta}_{t|t-1}^i$. ▷ by (22)
7: Calculate $q_t^i \propto p(\boldsymbol{y}_t^x|\boldsymbol{x}_t^i,\mu_t^i)$.
8: Draw indices $a_t^i \sim \mathcal{C}(\{q_t^i\}_{i=1}^N)$.
9: Resample $\{\boldsymbol{x}_t^i,\boldsymbol{\xi}_t^i,\boldsymbol{\Theta}_{t|t}^i\} \leftarrow \{\boldsymbol{x}_t^{a^i},\boldsymbol{\xi}_t^{a^i},\boldsymbol{\Theta}_{t|t}^{a^i}\}$.
10: **end for**

IV. Experimental Results

In this section, the proposed algorithm is validated with experimental data from a test vehicle. The test vehicle is a VW Golf GTE plug-in hybrid equipped with sensors as described in section II. The road condition sensors are mounted behind the two front tires. All data is collected via a CAN-bus, connected to a *Sirius-i* data-acquisition system from *Dewesoft*. Data is sampled with a frequency of 100 Hz.

We use a *Automotive Dynamic Motion Analyzer* (ADMA) model *eco+* from *Genesys Systems* as reference for the vehicle's inertial motion. The ADMA is coupled with a *Novatel* GPS-receiver and uses a high-quality inertial measurement unit to provide the vehicles reference position, velocity and orientation. The raw, uncorrected measurements of the GPS-receiver are also used as sensor signals in the proposed particle filter.

Experiments are conducted on a test track in Jeversen, Germany, owned by the *ZF Group*. A satellite image of the track is shown in Fig. 5. The area on the tracks western side is intended for breaking tests and has lanes with different types of pavement. From west to east these are concrete, cobblestone and asphalt. These lanes are equipped with a sprinkler system to keep the road wet.

The experiments to validate the tire-road friction estimation will consist of emergency breaking test on the western track. The vehicle will drive onto the marked lanes in Fig.5, arriving from north-east with a constant velocity of approximately 90 km/h. Once the vehicle is on the correct road patch, emergency breaking will be initiated until the vehicle reaches standstill. We perform two trials for each type of pavement.

Fig. 5. Satellite image of the test track from Google Maps. The lanes, intended for breaking tests are marked in red. Pavement types are concrete, cobblestone and asphalt from west to east.

To highlight the algorithms ability to learn tire-road friction and road condition at specific locations, the collected sufficient statistics are carried over from one trial to the next.

Fig. 6 shows the posterior probability density functions $p(\mu)$ for tire-road friction μ. Each column represents one of the three lanes asphalt, concrete and cobblestone. Rows show the posterior density after each trial, the first row shows the prior. Each probability density plot has an additional histogram for the collected data samples. The density function was calculated by first averaging the sufficient statistics over all particles, which results in one density per combination of pavement type P and road-weather condition W. These densities were then combined by a weighted sum over all environmental conditions using variable elimination. The state space model presented in Section II enables the quantification of uncertainties in tire-road friction with respect to environmental conditions, and the resulting distribution is multi-modal. Fig. 6 includes annotations for the dominant prior modes, which results mainly from the uncertainty in road weather condition. After the first trial, uncertainty in tire-road friction μ decreases significantly. With the information from air temperature S_T, road condition sensor S_W and precipitation S_P the road-weather condition can be estimated. As a result, modes for dry and snowy road are weakened. After the second trial uncertainty is reduced further, showing that the algorithm can utilize the environmental map to retain, recall and update location specific information related to environmental states, including tire-road friction.

Information is not only gathered through the environmental sensors, but also the vehicle dynamics in situations with sufficient slip like emergency breaking. Only proposals for tire-road friction μ, which sufficiently explain the vehicles movement as an effect of tire traction are retained. To better highlight this fact, Fig. 7 shows the scatter plot of proposals for tire-road friction μ for the second trial on cobblestone over time t. The sudden jump at approximately $0.5\,\mathrm{s}$ results from the vehicle driving from the dry asphalt of the north-eastern track onto the wet cobblestone lane. The environmental models spatial discretization into patches allows for a quick reaction to these changing conditions. The breaking maneuver starts at approximately $3.5\,\mathrm{s}$ and the vehicle reaches stand-still at approximately $9\,\mathrm{s}$. Particle variance decreases during breaking, showing that excitation of vehicle dynamics through sufficient slip is naturally detected. The resulting selection of particles through resampling is retained by recursively updating the sufficient statistics of tire-road friction as described in Section III-A and carried over into subsequent iterations. As a result, variance of the posterior density function also decreases, which shows in the generated samples. Comparing the distribution of particles in Fig. 7 before and after breaking shows that less particles are generated above a μ value of 0.75.

TABLE II
RMSE FOR PREDICTED BREAKING DISTANCE OF THE FIRST AND SECOND TRIAL OVER ALL PARTICLES.

Lane	Trial 1	Trial 2	Improvement
Asphalt	$4.37\,\mathrm{m}$	$3.98\,\mathrm{m}$	$9.80\,\%$
Concrete	$3.47\,\mathrm{m}$	$3.26\,\mathrm{m}$	$6.44\,\%$
Cobblestone	$10.94\,\mathrm{m}$	$10.90\,\mathrm{m}$	$0.37\,\%$
Average	$6.35\,\mathrm{m}$	$6.05\,\mathrm{m}$	$4.95\,\%$

To further quantify the estimation performance for tire road friction μ, we compare the proposals of the particle filter to the actual deceleration behavior of the vehicle. During breaking, the vehicles kinetic energy is dissipated through friction in the tire-road contact. Using energy conservation, we calculate the remaining distance to standstill (RDS)

$$s = \frac{1}{2}\frac{v_x^2}{\mu g}, \tag{35}$$

with longitudinal velocity v_x, gravitation g and tire-road friction μ. These values are compared against the actual distance at which the vehicle reaches standstill, taken from the ADMA reference. Fig.8 shows the calculated samples for RDS over time. Time was normalized to show all plots over a unified time scale. The time window was determined automatically, beginning when the acceleration drops below $-4\,\mathrm{m/s^2}$ and ending once the vehicle comes to a complete stop. In all trials, the predictions cluster around the true RDS, meaning the proposals for tire-road friction μ are generally plausible. To quantify the prediction error, we calculate the RMSE between the true RDS and the respective predictions at all time steps. The RMSE values are shown in Table II. For each lane, the RMSE of the second trial is smaller than the first, indicating lover overall variance and stronger clustering around true RDS and higher accuracy in predicting tire-road friction. On average, the prediction error between the first and second trial is reduced by $4.95\,\%$.

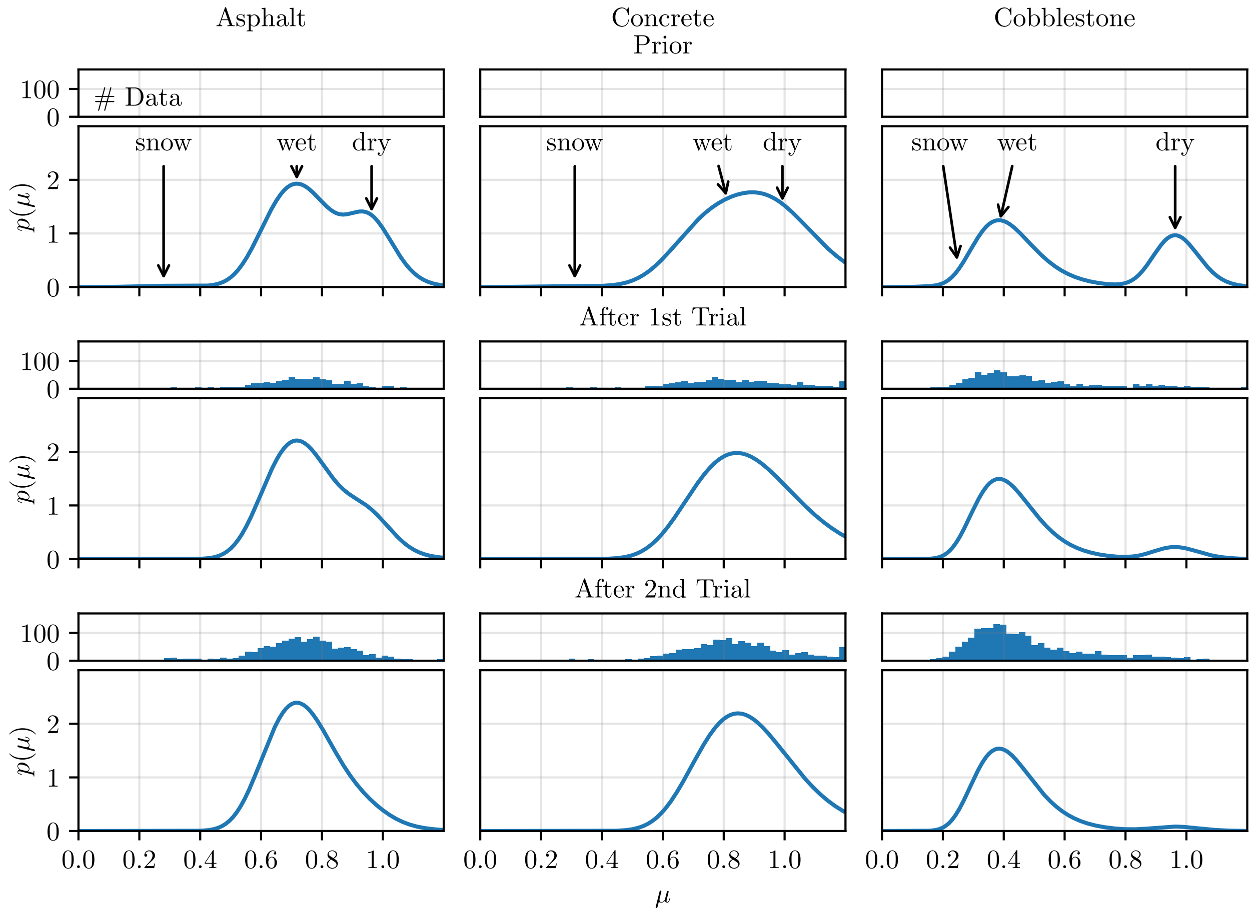

Fig. 6. Posterior probability density function $p(\mu)$ for tire-road friction μ, integrated over the probabilities of pavement type R and road weather condition W. Columns show the respective lanes of the test track. Rows are the resulting probability density function after each trial. The first row shows the prior. Histograms show the amount of collected data points. Annotations highlight modes resulting from specific road weather conditions.

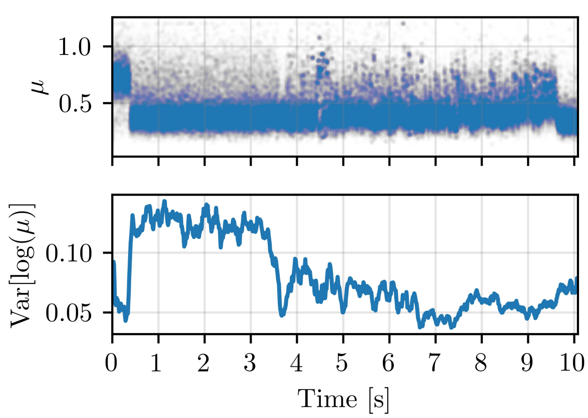

Fig. 7. Scatter plot of samples for tire-road friction μ over time t for the second trial on cobblestone (top row) and corresponding sample variance of $\log(\mu)$ over time (bottom row). Moving average smoothing was applied to the variance, to mage the graph better readable.

V. Conclusion

In this paper we presented a novel scheme for the estimation of tire road friction utilizing a Rao-Blackwellized particle filter. The structure of the proposed state space model represents the causal chain from causes of tire-road friction to the effects on vehicles movement, by combining a map of the local environment with a vehicle model. The proposed algorithm was validated on real data, by performing successive trials of emergency breaking on different types of pavement. By jointly estimating local environmental and vehicle states, we are able to quantify uncertainties in tire-road friction with respect to different environmental conditions. The environment is represented as a 2D map comprised of discrete road patches. Each patch can hold a different probability distribution for its environmental condition. Using a tailored Monte-Carlo scheme, the proposed algorithm is able to update the hyper-parameters of each individual patch online. In doing so, information for specific road patches can be saved and later recalled for future predictions. As a result, we achieve for the first time, 2D mapping of tire road friction in an online setting by combining cause- and effect based-methods. Results show, that uncertainty in tire-road friction for each patch of pavement can be reduced over successive breaking tests. The information about environmental states in each patch also enables a quick reaction to changing road conditions. Performance of tire-road friction estimation during emergency breaking was quantified by predicting the remaining distance to standstill. The results show that the prediction error between

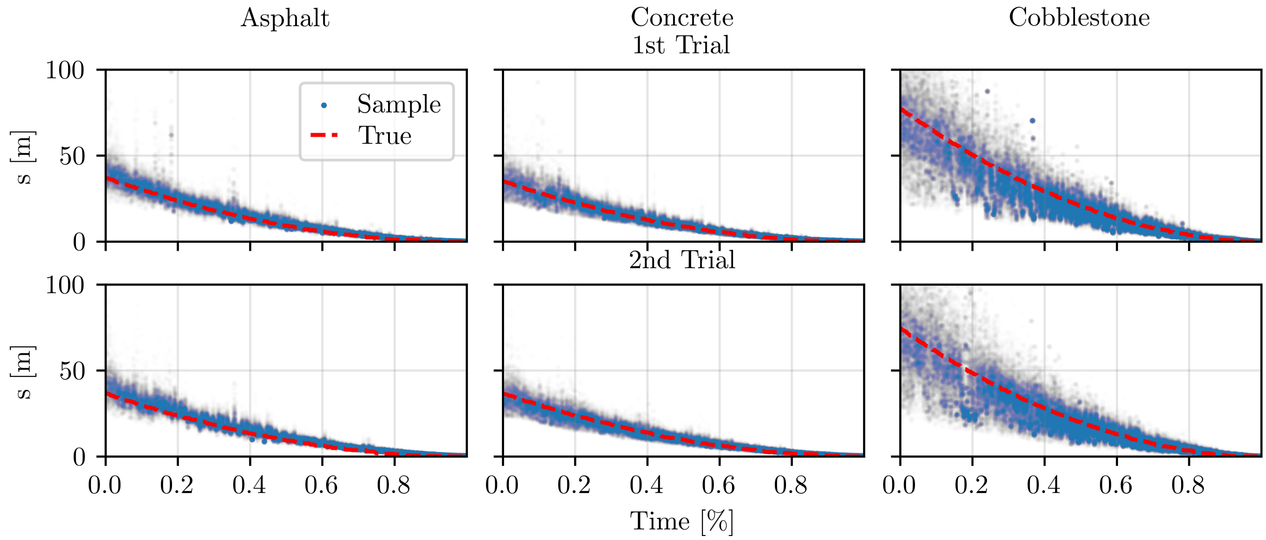

Fig. 8. Scatter plot of calculated RDS over normalized time. Blue dots are the calculated samples, true distance to standstill is shown in red.

two trials could be reduced by $4.95\,\%$ on average. This proves that in the future such a system could be used to enhance advanced driver assistance systems, increasing traffic safety as a result. Additionally, recalling of information for a specific road patch could enable autonomous vehicles or robots to perform more advanced task planning, leading to increased autonomy. Future research should be aimed at enabling the sharing of generated map information between vehicles. If vehicles would be able to share their experiences, the speed of exploration for a given region would be greatly increased. As a result, a vehicles predictions of tire road friction for a road patch could be increased even before the vehicle has ever visited that patch.

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Declaration of Interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

Vehicle Model

At time t the state of the vehicle

$$\boldsymbol{x}_t = [\boldsymbol{p}, \psi, \dot{\psi}, \omega_{1:4}, v_x, v_y]^{\mathrm{T}}$$

is described by position $\boldsymbol{p}$, yaw angle ψ, tire rotational rates $\omega_{1:4}$ for all four tires and velocities v_x and v_y in longitudinal and lateral direction respectively. It is assumed, that the vehicle drives on a horizontal plane. The dynamics of the vehicle are based on a two-track model with the origin of the body fixed frame being on the center of gravity and x- and z-axis pointing to the front and up respectively. A sketch of the model is given in Fig.9.

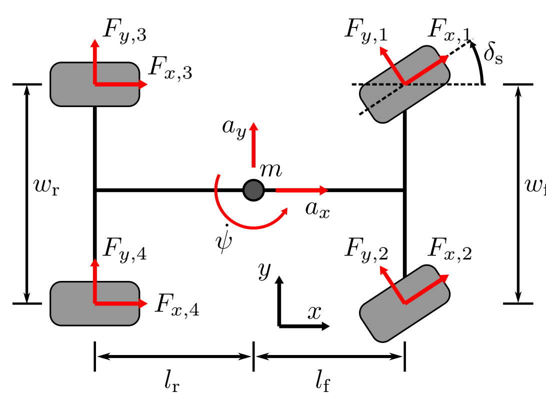

Fig. 9. Sketch of the two-track model, with side view (top) and birds-eye view (bottom). The center of gravity is marked by point mass m.

The vertical static tire loads $F_{z,1:4}$ are calculated by

$$\begin{aligned} m_f &= \frac{m l_{\mathrm{r}}}{l_{\mathrm{f}} + l_{\mathrm{r}}}, & m_r &= \frac{m l_{\mathrm{f}}}{l_{\mathrm{f}} + l_{\mathrm{r}}}, \\ F_{z,1} = F_{z,2} &= \frac{m_{\mathrm{f}} g}{2}, & F_{z,3} = F_{z,4} &= \frac{m_{\mathrm{r}} g}{2}, \end{aligned} \tag{36}$$

with virtual masses m_{f} and m_{r} for the front an rear axle respectively, total vehicle mass m and gravitation g. Small shifts in the position of the center of gravity by pitch and roll movement are neglected.

Tire-slip is calculated with

$$s_{x,i} = \frac{r\omega_i - v_{x,i,\mathrm{t}}}{\max(v_{x,i,\mathrm{t}}, v_{\min})}, \quad s_{y,i} = -\frac{v_{y,i,\mathrm{t}}}{\max(v_{x,i,\mathrm{t}}, v_{\min})}, \tag{37}$$

where $v_{x,i,\mathrm{t}}$ and $v_{y,i,\mathrm{t}}$ are longitudinal and lateral velocity with respect to tire i and $v_{\min}$ is the lower bound for the tires longitudinal velocity used for normalization. With a positive lower bound $v_{\min}$ the slip calculation avoids division by zero. Additionally, lower bound $v_{\min}$ is also used as a trade-off between model accuracy and computational cost. This is because the standard normalization of slip results in stiff differential equations at lower velocities, which requires shorter time steps for the solver.

Calculation of the tire-road contact forces is done with the Magic-Tire model [40] by

$$\begin{aligned}
F_* &= \frac{s_*}{s} D \sin(C \arctan(B s_* - E(B s_* - \arctan(B s_*)))),\\
C_F &= c_1 \sin(2 \arctan(F_z / c_2)),\\
D &= \mu F_z,\\
B &= \frac{C_F}{CD},\\
s &= \sqrt{s_x^2 + s_y^2}
\end{aligned} \tag{38}$$

with s_* being either longitudinal or lateral slip. Here, we use the same parameters for all tires, but with different characteristics in x and y direction.

The equations of motion are formulated by Newton-Euler formalism. For velocity, the differential equation is defined as

$$\begin{aligned}
\dot{v}_x &= \frac{1}{m} \left[\sum_{i=1}^{4} F_{x,i} - F_{\text{air}} \right] + \dot{\psi} v_y - g_x\\
\dot{v}_y &= \frac{1}{m} \left[\sum_{i=1}^{4} F_{y,i} \right] - \dot{\psi} v_x - g_y\\
F_{\text{air}} &= \frac{1}{2} \rho_{\text{air}}\, C_d\, A_x\, v_x^2\, \text{sign}(v_x)
\end{aligned} \tag{39}$$

with $F_{x,i}$ and $F_{y,i}$ as the tire contact forces from equation (38) and F_{air} as air resistance. For the air resistance F_{air}, ρ is the air density, C_d is the air resistance coefficient and A_x is the front surface of the vehicle. The sign is used to account for forward and backward driving.

The dynamic of yaw movement is calculated with the sum of moments by

$$\begin{aligned}
\ddot{\psi} = &\frac{1}{I_z}[(F_{y,1} + F_{y,2}) l_f - (F_{y,3} + F_{y,4}) l_r] +\\
&\frac{1}{I_z}[(F_{x,1} - F_{x,2}) \frac{w_f}{2} + (F_{x,3} - F_{x,4}) \frac{w_r}{2}],
\end{aligned} \tag{40}$$

with I_z as the vehicles inertia around the z-axis.

Dynamics for the tire angular velocity ω are calculated by

$$\begin{aligned}
\dot{\omega} &= \frac{T_{\text{d}} - T_{\text{r}}}{I},\\
T_{\text{d}} &= T_{\text{e}} i_{\text{g}},\\
T_{\text{r}} &= F_x r + F_r + T_{\text{b}} \text{sign}(\omega),\\
F_{\text{r}} &= F_z (c_{\text{r},1} + c_{\text{r},2} |\omega|) \text{sign}(\omega),\\
I &= I_{\text{t}} + \frac{1}{2} I_{\text{e}}\, i_{\text{g}}^2,
\end{aligned} \tag{41}$$

with engine torque T_{e}, engine inertia I_{e}, break torque T_{b}, gear transmission i_{g}, tire radius r, tire inertia I_{t} and rolling resistance F_{r}.

Since we assume driving on a horizontal plane, the position is represented using local x and y coordinates. The local navigation frame n is placed on a reference point in geodetic coordinates, tangential to the earth's surface. Here, we use the ENU convention, meaning the x-axis points east and the y-axis towards north. A given geodetic location $[\phi, \lambda, h]$ in latitude ϕ, longitude λ and height h can be converted to earth centered, earth fixed (ECEF) coordinates by

$$\begin{aligned}
X &= (N(\phi) + h) \cos\phi \cos\lambda,\\
Y &= (N(\phi) + h) \cos\phi \sin\lambda,\\
Z &= \left[\left(1 - e^2\right) N(\phi) + h \right] \sin\phi,\\
N(\phi) &= \frac{a}{\sqrt{1 - e^2 \sin^2\phi}},\\
e &= 1 - \frac{b^2}{a^2}
\end{aligned} \tag{42}$$

with a and b as the reference ellipsoids equatorial and polar radius respectively. Applying equation (42) to a reference point $[\phi_0, \lambda_0, h_0]$ and and a target point $[\phi, \lambda, h]$, the difference in ECEF coordinates is calculated by

$$\Delta X = X - X_0, \quad \Delta Y = Y - Y_0, \quad \Delta Z = Z - Z_0. \tag{43}$$

Rotating this difference into the ENU frame yields the local coordinates

$$\begin{bmatrix} E \\ N \\ U \end{bmatrix} = \begin{bmatrix} -\sin\lambda_0 & \cos\lambda_0 & 0 \\ -\sin\phi_0 \cos\lambda_0 & -\sin\phi_0 \sin\lambda_0 & \cos\phi_0 \\ \cos\phi_0 \cos\lambda_0 & \cos\phi_0 \sin\lambda_0 & \sin\phi_0 \end{bmatrix} \begin{bmatrix} \Delta X \\ \Delta Y \\ \Delta Z \end{bmatrix}. \tag{44}$$

Because the vehicle only moves in a horizontal plane, we have 2D coordinates

$$\begin{aligned}
\boldsymbol{p} &:= \begin{bmatrix} E \\ N \end{bmatrix},\\
\dot{\boldsymbol{p}} &= \begin{bmatrix} \cos\psi & -\sin\psi \\ \sin\psi & \cos\psi \end{bmatrix} \begin{bmatrix} v_x \\ v_y \end{bmatrix}.
\end{aligned} \tag{45}$$

Using the above equations, we have the vehicles state dynamics as

$$\dot{\boldsymbol{x}}_t = [\dot{\boldsymbol{p}}, \dot{\psi}, \ddot{\psi}, \dot{\omega}_{1:4}, \dot{v}_x, \dot{v}_y]^{\text{T}}. \tag{46}$$

The time discrete state space model for the vehicle can be formulated by applying a suitable solver to equation (46). Here, we use the Euler method with M intermediary steps

$$\begin{aligned}
\boldsymbol{x}_{t+1} := \boldsymbol{x}_{t,M} &= \sum_{i=0}^{M} \boldsymbol{x}_{t,i} + \Delta t\, \dot{\boldsymbol{x}}_{t,i}\\
\boldsymbol{x}_{t,0} &:= \boldsymbol{x}_t\\
\dot{\boldsymbol{x}}_{t,0} &:= \dot{\boldsymbol{x}}_t
\end{aligned} \tag{47}$$

to achieve a higher effective sample frequency $\Delta t / M$, for numerically stable integration. Solving the vehicle dynamics at lower velocities (e.g. breaking to standstill) requires smaller integration steps due to the stiffening of the differential equations. More advanced solvers like Runge-Kutta-4 were considered, but in comparison, Euler yields a stable solution with the least amount of function evaluations and negligible differences in the calculated solutions. The parameters of the model are given in Table III.

TABLE III
PARAMETERS FOR THE VEHICLE DYNAMICS MODEL

Parameter	Value	SI Unit
c_{1x}	2.5×10^7	$\mathrm{N\,rad^{-1}}$
c_{2x}	3.6×10^6	N
c_{1y}	1.9×10^6	$\mathrm{N\,rad^{-1}}$
c_{2y}	1.35×10^5	N
C_x	1.42	-
C_y	1.9	-
E_x	-9.75	-
E_y	0.52	-
$c_{\mathrm{r},1}$	0.0083	-
$c_{\mathrm{r},2}$	0.0005	-
m	1720	kg
I_z	2066	$\mathrm{kg\,m^2}$
I_{t}	28.6	$\mathrm{kg\,m^2}$
I_{e}	0.197	$\mathrm{kg\,m^2}$
C_{d}	0.27	-
w_f	1.543	m
w_r	1.513	m
l_f	1.16	m
l_r	1.47	m
r_t	0.3116	m
$v_{\min}$	1	$\mathrm{m\,s^{-1}}$
g	9.81	$\mathrm{m\,s^{-2}}$
ρ_{air}	1.225	$\mathrm{kg\,m^{-3}}$
A_x	2.19	$\mathrm{m^2}$
Δt	0.01	s
M	20	-

ENVIRONMENT MODEL

To reiterate, we define the environmental condition as

$$\boldsymbol{\xi}_t = [T, P, W, R, \mu]^{\mathrm{T}},$$

with pavement temperature T, precipitation P, road-weather condition W, road-pavement type R and the maximum tire-road friction potential μ. Information about the local environment is gathered via measurements from an air temperature sensor S_T, two piezo acoustic sensors S_W for detection of water on the roads surface and a rain sensor S_P. As detailed in section II, the models parameters are the hyper-parameters of the respective variables conjugate prior distribution $\boldsymbol{\Theta}$, as well as conditional probabilities for measurement values. For the Dirichlet distribution, these hyper-parameters are the absolute number of occurrences $\boldsymbol{z}$ for observed classes. To define this prior, we will use relative occurrences $\tilde{\boldsymbol{z}}$ which can be scaled by

$$\boldsymbol{z} = \tilde{\boldsymbol{z}}\, n, \tag{48}$$

using a virtual number of prior observations n. Parametrization is partially based on our previous work [35], [36].

Informed priors for road patches

The network topology shown in Fig 2 is build for each road patch j individually by selecting the respective parameters from $\boldsymbol{\theta}$. Previously, without spatial resolution, the root nodes T, P and R were given a static uniform priors to avoid bias of the estimation. With added the spatial resolution, it is now possible to give each patch an informed prior for root nodes T, P and R.

Experimental validation is performed with data recorded on a test track in Jeversen Germany, owned by the ZF Group. Patch definition is done for the specific pavements of this test track, which are

- asphalt,
- concrete,
- cobblestone
- rest.

The eastern side of the track is equipped with a sprinkler system to simulate rain and keep the road surface wet. For this reason, we distinguish between the asphalt on the eastern side and the rest of the track specifically. Because the premise is privately owned, the priors for each patch are defined manually to emulate the information received from corresponding online providers like *OpenStreetMap* [41].

Following from the selected patches, we distinguish between asphalt, cobblestone and concrete for pavement type R. The corresponding parameters are given in Table IV. These values were chosen to mimic the accuracy of *OpenStreetMap* as used in [36].

TABLE IV
PARAMETERS $\tilde{\boldsymbol{Z}}_R(j)$ FOR PAVEMENT TYPE R.

Patch	Asphalt	Concrete	Cobblestone
asphalt	90 %	5 %	5 %
concrete	5 %	90 %	5 %
cobblestone	5 %	5 %	90 %
rest	90 %	5 %	5 %

Precipitation P is a binary value of *true* or *false*. The corresponding parameters are given in Table V. Because the eastern patches are equipped with a sprinkler system, they receive a higher likelihood of rain.

TABLE V
PARAMETERS $\tilde{\boldsymbol{Z}}_P(j)$ FOR PRECIPITATION P.

Patch	*false*	*true*
asphalt	10 %	90 %
concrete	10 %	90 %
cobblestone	10 %	90 %
rest	90 %	10 %

Pavement temperature T is defined as discrete intervals

- $T_1 : T < -21\,°\mathrm{C}$
- $T_2 : -21\,°\mathrm{C} \leq T < -15\,°\mathrm{C}$
- $T_3 : -15\,°\mathrm{C} \leq T < -10\,°\mathrm{C}$
- $T_4 : -10\,°\mathrm{C} \leq T < -5\,°\mathrm{C}$
- $T_5 : -5\,°\mathrm{C} \leq T < 0\,°\mathrm{C}$
- $T_6 : 0\,°\mathrm{C} \leq T$

The lower bound is chosen as $-21\,°\mathrm{C}$, because it is the freezing temperature of saturated saline solution, while salt is often utilized to prevent water on roads from freezing. Due to the close proximity of all patches, the same prior is used for each patch. The probabilities for each class are calculated from data of the *Iowa State University Environmental Mesonet*[1]. The measurements were taken from the US states Missury and New Hampshire, because their average temperatures are close to the climate of Germany. The values are given in Table VI.

[1]The data can be accessed under https://mesonet.agron.iastate.edu/request/download.phtml

TABLE VI
PARAMETERS $\tilde{\boldsymbol{Z}}_T(j)$ FOR PAVEMENT TEMPERATURE T.

Patch	T_1	T_2	T_3	T_4	T_5	T_6
any	0.00 %	0.00 %	0.41 %	3.09 %	8.93 %	87.57 %

TABLE VII
PARAMETERS $\tilde{\boldsymbol{Z}}_W(T, P)$ FOR ROAD-WEATHER CONDITION W.

T	P	*dry*	*wet*	*snow*
T_1	False	97.44 %	0.00 %	2.56 %
T_1	True	9.52 %	0.00 %	90.48 %
T_2	False	97.44 %	0.51 %	2.05 %
T_2	True	9.52 %	18.10 %	72.38 %
T_3	False	97.44 %	1.03 %	1.54 %
T_3	True	9.52 %	36.19 %	54.29 %
T_4	False	97.44 %	1.54 %	1.03 %
T_4	True	9.52 %	54.29 %	36.19 %
T_5	False	97.44 %	2.05 %	0.51 %
T_5	True	9.52 %	72.38 %	18.10 %
T_6	False	97.44 %	2.56 %	0.00 %
T_6	True	9.52 %	90.48 %	0.00 %

A convenient way to infer patch-specific class probabilities $\boldsymbol{B}_T(j)$ and $\boldsymbol{B}_P(j)$ for pavement temperature and precipitation in a real world setting is to employ cloud-based road weather models by commercial providers such as *Klimator* (*Road Cloud Data* [42]) or *Vaisala* (*Xweather*). These services have their roots in mechanistic road-condition forecast systems like METRo [43] and typically come in the form of APIs that generate pavement temperature estimates and precipitation probabilities at a given geolocation.

Road-weather condition

For the road-weather condition, we distinguish between the classes

- dry,
- wet,
- snow.

The prior only depends on precipitation P and pavement temperature T and is defined independent of the road patches. The parameters are defined such that, in case of precipitation, regardless of temperature, the total probability of wet or snowy road is 95 %. For temperature, the probability that water on the roads surface is frozen decreases 20 % for each temperature class, until 0 % for T_6.

Tire-road friction potential

The tire-road friction potential μ depends on road-weather condition and pavement type. The corresponding conjugate prior is a normal-inverse-gamma distribution. We define one set of parameters for each combination of conditional variables. Here, we use data collected by the *TU Berlin* [44] to initialize the prior sufficient statistics for tire road friction. The data contains samples of tire-road friction for different combination of pavement and road-weather condition, which were gathered around Berlin, Germany.

TABLE VIII
PARAMETERS $\boldsymbol{B}_{S_T}(T)$ FOR THE AIR TEMPERATURE SENSOR S_T.

T	$S_{T,1}$	$S_{T,2}$	$S_{T,3}$	$S_{T,4}$	$S_{T,5}$
T_1	99.97 %	0.03 %	0.00 %	0.00 %	0.00 %
T_2	97.57 %	2.42 %	0.01 %	0.00 %	0.00 %
T_3	69.11 %	29.28 %	1.60 %	0.01 %	0.00 %
T_4	18.53 %	58.43 %	22.21 %	0.83 %	0.00 %
T_5	1.05 %	24.58 %	58.05 %	15.93 %	0.38 %
T_6	0.00 %	0.23 %	3.97 %	11.27 %	84.54 %

Air Temperature

Sensors for air temperature can be considered standard on modern vehicles. Air temperature is divided into the discrete intervals

- $S_{T,1} : T_{\text{air}} < -10\,°\text{C}$
- $S_{T,2} : -10\,°\text{C} \leq T_{\text{air}} < -5\,°\text{C}$
- $S_{T,3} : -5\,°\text{C} \leq T_{\text{air}} < 0\,°\text{C}$
- $S_{T,4} : 0\,°\text{C} \leq T_{\text{air}} < 5\,°\text{C}$
- $S_{T,5} : 5\,°\text{C} \leq T_{\text{air}}$

Similar to [35], probabilities of S_T given T are calculated from data of the *Iowa State University Environmental Mesonet*. The parameters are given in Table VIII. Similar to the prior for surface temperature, these values are based on measurements from the US states Missury and New Hampshire.

Piezo acoustic sensor

The vehicle is equipped with two piezo-acoustic road condition sensors (RCS), manufactured by *HELLA GmbH & Co. KGaA*, an automotive supplier. They are mounted radial in the wheel house behind the two front wheels of the vehicle. The sensor uses a piezo-electric surface to detect the impacts of water drops whirled up by the tire's rotation. The resulting electric signals are processed inside the sensor and converted into a level of wetness, which is a 16 bit integer. The RCS classifies the road weather condition as level of wetness. The sensor signal is discretized into 3 classes. The distribution was summarized, adapted and quantified in cooperation with the manufacturer, based on data recorded with the available test vehicle. The probability distribution is shown in Table IX.

TABLE IX
PARAMETERS $\boldsymbol{B}_{S_W}(W)$ FOR THE ROAD CONDITION SENSOR S_W.

W	$S_{W,1}$	$S_{W,2}$	$S_{W,3}$
dry	95.00 %	5.00 %	0.00 %
wet	17.50 %	26.25 %	56.25 %
snow	96.00 %	4.00 %	0.00 %

Rain sensor

The rain sensor is part of the vehicles standard equipment and mounted behind windshield, next to the back mirror. The sensor returns the probability of precipitation in percent, so the corresponding parameters can be calculated online with

$$\boldsymbol{B}_{S_P}(S_P) = \frac{1}{100}\begin{bmatrix}100 - S_P \\ S_P\end{bmatrix}. \tag{49}$$

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